A Unified Theory of Bounded Dual Valuation

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Abstract

We study a pair of real-valued functions $f(x,y)$ and $g(x,y)$ satisfying $fg = r^2$ and $f(0,0) = g(0,0) = c^2$, together with the bounded-sum condition $m \le f + g \le M$. Combining the submitted frameworks of *Unification of Analysis*, *Blancmange Analysis (Stronger Version)*, and *Arbitrage Geometry*, we show that $r^2 = c^4$, that the positive branch admits the one-field representation $f = c^2 e^h$ and $g = c^2 e^{-h}$, and that the additive bound controls both the latent field h and the valuation discrepancy $f - g$. Stronger Blancmange Analysis supplies a dyadic fold metric, fold derivatives, and Faber–Schauder-type localized expansions; Arbitrage Geometry interprets f and g as competing valuations and identifies the same discrepancy as an arbitrage exposure. The paper then records concise implications for economics, finance, econometrics, statistics, computer science, data science, artificial intelligence, and selected physical, biological, psychological, philosophical, political, and strategic analogies. These latter interpretations are presented as conceptual mappings rather than empirical laws.

The paper ends with "The End"

1 Problem and Unified Representation

Let

$$f(x,y)g(x,y) = r^2, \qquad f(0,0) = g(0,0) = c^2, \tag{1}$$

where $r, c \in \mathbb{R}$. Evaluating the first relation at the origin gives

$$r^2 = c^4. \tag{2}$$

Hence

$$f(x,y)g(x,y) = c^4. \tag{3}$$

Impose additionally

$$m \le f(x,y) + g(x,y) \le M. \tag{4}$$

For the positive branch suggested by the common value $c^2 > 0$, write

$$f(x,y) = c^2 e^{h(x,y)}, \qquad g(x,y) = c^2 e^{-h(x,y)}. \tag{5}$$

Then

$$f + g = 2c^2 \cosh h, \qquad f - g = 2c^2 \sinh h. \tag{6}$$

Therefore

$$m \le 2c^2 \cosh h(x,y) \le M. \tag{7}$$

Provided $M \ge 2c^2$, the upper bound implies

$$|h(x,y)| \le \operatorname{arcosh}\left(\frac{M}{2c^2}\right). \tag{8}$$

The reciprocal system also satisfies

$$(f+g)^2-(f-g)^2=4c^4, \tag{9}$$

so that

$$|f-g|=\sqrt{(f+g)^2-4c^4}\le\sqrt{M^2-4c^4}. \tag{10}$$

The minimum possible sum on the positive branch is $2c^2$, by the arithmetic–geometric mean inequality, with equality if and only if $f=g=c^2$.

$$2c^2\le f+g\le M,\qquad 0\le|f-g|\le\sqrt{M^2-4c^4}. \tag{11}$$

2 Blancmange Analysis and Fold Geometry

The stronger Blancmange framework replaces or supplements ordinary infinitesimal descent by canonical dyadic scales 2^{-n} and uses the Takagi–Landsberg family

$$B_\lambda(x)=\sum_{n=0}^{\infty}\lambda^n\psi(2^nx),\qquad \psi(x)=\operatorname{dist}(x,\mathbb{Z}),\qquad 0<\lambda<1, \tag{12}$$

whose continuity follows from uniform convergence [2,3]. The associated fold metric is

$$d_{\lambda,\eta}(x_1,x_2)=|x_1-x_2|+\eta|B_\lambda(x_1)-B_\lambda(x_2)|,\qquad \eta>0. \tag{13}$$

It is a metric and induces the ordinary topology on the interval [3].

The dyadic fold address is a binary sequence $(\epsilon_n)_{n\ge1}$ with

$$x=\sum_{n=1}^{\infty}\frac{\epsilon_n}{2^n},\qquad \epsilon_n\in\{0,1\}, \tag{14}$$

modulo the standard two-address identification at dyadic rationals [3]. Thus the latent field h may be studied through its behavior under nested dyadic descent rather than only through Euclidean increments.

For admissible points, define the fold quotient and fold derivative by

$$Q_n^{B_\lambda}h(x)=\frac{h(x+2^{-n})-h(x)}{B_\lambda(x+2^{-n})-B_\lambda(x)}, \tag{15}$$

$$D_{B_\lambda}h(x)=\lim_{n\to\infty}Q_n^{B_\lambda}h(x), \tag{16}$$

whenever the indicated limit exists [3]. The localized approximation system should use genuine dyadic hat functions $\phi_{n,k}$ rather than periodic shifts of ψ; these Faber–Schauder-type atoms are uniformly dense in $C[0,1]$ [3,4]. Accordingly, on a one-dimensional slice of the (x,y) domain one may write the latent field schematically as

$$h_N(x)=a_0+a_1x+\sum_{n=0}^{N}\sum_{k=0}^{2^n-1}c_{n,k}\phi_{n,k}(x). \tag{17}$$

The entire pair is then generated from one coefficient field:

$$f_N(x)=c^2e^{h_N(x)}, \tag{18}$$

$$g_N(x)=c^2e^{-h_N(x)}. \tag{19}$$

This is preferable to estimating f and g independently when the product restriction is treated as structural.

A useful weighted fold-complexity functional is

$$C_s(h_N) = \sum_{n=0}^{N} 2^{2sn} \sum_{k=0}^{2^n-1} c_{n,k}^2, \qquad s \geq 0, \tag{20}$$

which penalizes increasingly fine dyadic structure [3].

3 Arbitrage Geometry of the Reciprocal Pair

Arbitrage Geometry represents a system as

$$\mathcal{A} = (X, \mathcal{T}, \mathcal{V}, C), \tag{21}$$

where X is the state space, $\mathcal{T}$ admissible transformations, $\mathcal{V}$ a family of valuations, and C transformation cost [5]. The central arbitrage functional is

$$\Pi(T, z; i, j) = V_j(Tz) - V_i(z) - C(T). \tag{22}$$

Set

$$V_f(z) = f(z), \qquad V_g(z) = g(z), \qquad z = (x, y). \tag{23}$$

For the identity transformation I,

$$\Pi(I, z; f, g) = g(z) - f(z) - C(I), \tag{24}$$
$$\Pi(I, z; g, f) = f(z) - g(z) - C(I). \tag{25}$$

Hence, if valuation switching is costless, $C(I) = 0$, global no-arbitrage requires

$$f(z) = g(z). \tag{26}$$

Combining this with $fg = c^4$ yields

$$f(z) = g(z) = c^2. \tag{27}$$

Thus the symmetric point is simultaneously the product-consistent equilibrium and the costless-switching no-arbitrage state.

If switching carries a constant cost $\kappa \geq 0$, both directions are nonprofitable precisely when

$$|f(z) - g(z)| \leq \kappa. \tag{28}$$

Since

$$|f(z) - g(z)| \leq \sqrt{M^2 - 4c^4}, \tag{29}$$

a sufficient global no-arbitrage condition is

$$\kappa \geq \sqrt{M^2 - 4c^4}. \tag{30}$$

This gives an operational interpretation of the additive bound: it controls the largest valuation misalignment that can be extracted by a pure switching transformation.

4 Local Arbitrage Gradients and Fold Dynamics

Where ordinary differentiability exists, the arbitrage gradient for the ordered valuation pair (f, g) is

$$\nabla_A^{fg}(z) = \nabla g(z) - \nabla f(z). \tag{31}$$

Using $f = c^2 e^h$ and $g = c^2 e^{-h}$ gives

$$\nabla f = f \nabla h, \qquad \nabla g = -g \nabla h, \tag{32}$$

and therefore

$$\boxed{\nabla_A^{fg}(z) = -2c^2 \cosh(h(z)) \nabla h(z).} \tag{33}$$

Thus the local direction of valuation discrepancy is determined by the gradient of the latent field, with magnitude scaled by the additive invariant.

For a fold-observable representation, the stronger Blancmange chain rule implies the analogous fold-relative relation whenever the fold derivative exists [3]. In particular, if $h = H(B_\lambda)$, then

$$D_{B_\lambda} f = c^2 e^h D_{B_\lambda} h = f D_{B_\lambda} h, \tag{34}$$

$$D_{B_\lambda} g = -c^2 e^{-h} D_{B_\lambda} h = -g D_{B_\lambda} h, \tag{35}$$

so that

$$D_{B_\lambda} g = -\frac{g}{f} D_{B_\lambda} f. \tag{36}$$

This is the fold-relative counterpart of $g\,df + f\,dg = 0$.

5 Cycle Consistency and Arbitrage Flatness

For a closed transformation path γ, Arbitrage Geometry defines the curvature-like quantity

$$\Omega(\gamma) = \sum_{k=1}^{m} \left[V_{i_k}(z_k) - V_{i_{k-1}}(z_{k-1}) - C(T_k) \right]. \tag{37}$$

Positive curvature means a net positive gain around the loop; zero curvature means an arbitrage-flat loop [5]. If every closed loop has zero total arbitrage weight on a connected transformation graph, the No-Arbitrage Flatness Theorem yields a global potential P satisfying

$$w(e) = P(b) - P(a) \tag{38}$$

for every edge $e : a \to b$ [5]. In the present setting, fold-scale movements and valuation switches may therefore be treated jointly: Blancmange structure specifies the multiscale geometry, while Arbitrage Geometry tests whether transformations through that geometry generate exploitable inconsistency.

6 A Compact Synthesis

The full system can be summarized by

$$f = c^2 e^h, \qquad g = c^2 e^{-h}, \tag{39}$$

$$fg = c^4, \qquad f + g = 2c^2 \cosh h, \tag{40}$$

$$f - g = 2c^2 \sinh h, \qquad (f+g)^2 - (f-g)^2 = 4c^4, \tag{41}$$

$$m \le f + g \le M, \qquad |h| \le \operatorname{arcosh}\left(\frac{M}{2c^2}\right), \tag{42}$$

$$|f - g| \le \sqrt{M^2 - 4c^4}, \tag{43}$$

with the last two bounds interpreted on the positive branch when $M \geq 2c^2$.

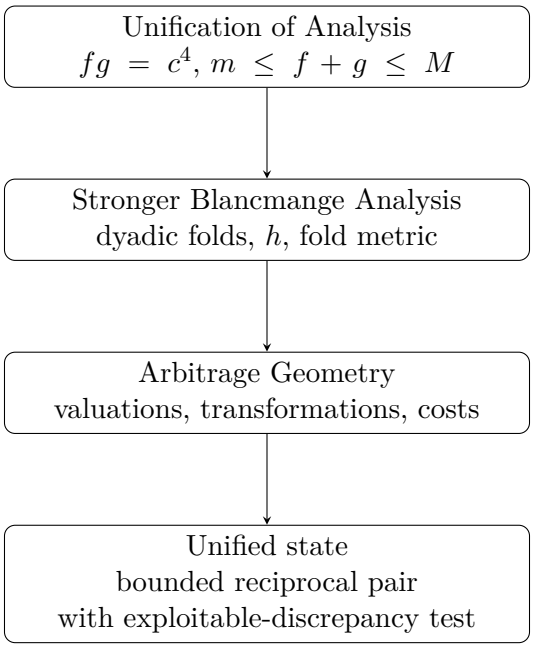


Figure 1: Structural synthesis of the three frameworks.

7 Implications for Economics, Finance, and Econometrics

The representation $f = c^2 e^h$, $g = c^2 e^{-h}$ is naturally interpreted as two reciprocal valuation indices around a reference level c^2. The sum $f+g$ measures aggregate level while $f-g$ measures asymmetry. This permits a compact econometric decomposition in which h is a latent index and the observed pair (f, g) is a nonlinear measurement system. In a statistical implementation, one may estimate h by localized tent regression, impose the product restriction exactly through exponentiation, and regularize fine-scale coefficients through C_s [3].

In finance, the two valuations may represent two venues, pricing conventions, risk-adjusted values, or state-contingent valuation maps. The arbitrage margin becomes a transformation-adjusted discrepancy, consistent with the no-arbitrage tradition [5,6]. In welfare economics, distinct valuations can instead represent utility or welfare criteria; the framework then separates valuation conflict from the cost of moving between allocations or institutions [7]. In macroeconomics and econometrics, h can be treated as an index of relative valuation pressure, while the bounded sum provides a state-space restriction.

Table 1: Selected economic and quantitative interpretations

Field	Object	Interpretation
Microeconomics	f, g	competing valuation or utility indices
Welfare economics	$f - g$	welfare-criterion disagreement
Finance	Π	cost-adjusted valuation arbitrage
Asset pricing	f, g	alternative state prices or valuation maps
Econometrics	h	latent nonlinear index
Statistics	$C_s(h)$	multiscale regularization
Time series	2^{-n} scales	dyadic multiscale decomposition

8 Statistics, Computer Science, Data Science, and AI

The stronger Blancmange framework supplies localized tent features, positive semidefinite fold kernels, and fold-regularized learning objectives [3]. A concise supervised-learning model is

$$\widehat{h}(x) = \theta^\top \Phi_N(x), \tag{44}$$

where Φ_N contains 1, x, the Takagi–Landsberg observable, and localized dyadic atoms. The induced predictors are

$$\widehat{f}(x) = c^2 e^{\widehat{h}(x)}, \tag{45}$$

$$\widehat{g}(x) = c^2 e^{-\widehat{h}(x)}. \tag{46}$$

This construction enforces the multiplicative invariant by design rather than as a soft penalty.

A fold-aware empirical objective may combine prediction loss with a fold-metric regularizer,

$$\mathcal{L}(\theta) = \frac{1}{N_s}\sum_{i=1}^{N_s} \ell\left(y_i, F_\theta(z_N(x_i))\right) + \rho \sum_{i,j} \frac{|F_\theta(z_N(x_i)) - F_\theta(z_N(x_j))|^2}{d_{\lambda,\eta}(x_i,x_j)^2 + \delta}, \tag{47}$$

following the stronger framework [3].

Algorithmic template

Input: (x_i, y_i), $c, m, M, \lambda, \eta, N$.
Step 1: construct dyadic fold features and localized hats $\phi_{n,k}$.
Step 2: estimate $\widehat{h}$ by least squares or penalized regression.
Step 3: clip or reject states violating $2c^2 \le 2c^2 \cosh \widehat{h} \le M$. (48)
Step 4: $\widehat{f} = c^2 e^{\widehat{h}}, \quad \widehat{g} = c^2 e^{-\widehat{h}}$.
Step 5: compute $\widehat{A} = \widehat{f} - \widehat{g}$ and transformation gains $\widehat{\Pi}$.
Step 6: search for positive-gain edges and positive-curvature cycles.

This template combines the approximation and learning procedures described in the stronger Blancmange framework with the exhaustive and cycle-based arbitrage detection procedures of Arbitrage Geometry [3,5].

9 Cross-Disciplinary Interpretation

The same mathematical skeleton can be used as a comparative language across disciplines, provided that one distinguishes structural analogy from established empirical theory. In physics, h can be viewed formally as a log-ratio coordinate between two conjugate measurements; the relation $(f+g)^2-(f-g)^2 = 4c^4$ has the algebraic form of an invariant hyperbola, but no physical law is asserted. In chemistry, the pair may represent two normalized activities or competing effective valuations subject to a conservation-like product constraint. In biology, h can represent a latent log-odds or relative-state index in a stylized two-state system. In psychology and psychiatry, the construction may serve as a mathematical analogy for competing appraisals or reciprocal latent tendencies, but it is not a diagnostic or clinical model. In philosophy, the pair illustrates how dual descriptions can share an invariant while remaining locally asymmetric. In religion, the zero-field state $h = 0$ may be read metaphorically as a state of balance, without making theological claims.

In political science, geopolitics, diplomacy, and international relations, f and g can be interpreted as competing institutional or strategic valuations. In warfare economics, Π measures whether a transformation converts asymmetry into net advantage after costs. Such uses remain

formal extensions of the valuation-and-transformation language of Arbitrage Geometry rather than empirical strategic doctrines [5].

Table 2: Conceptual mappings beyond economics

Field	Formal analogue	Caution
Physics	invariant and conjugate coordinates	algebraic analogy only
Chemistry	competing activities under constraint	requires domain-specific thermodynamics
Biology	latent relative-state field h	requires biological observables
Psychology	competing appraisal indices	not a clinical model
Psychiatry	reciprocal latent scores	not diagnostic
Philosophy	dual description with invariant	interpretive, not a proof of doctrine
Religion	balance at $h = 0$	metaphorical only
Political science	institutional valuation gap	requires institutional data
Geopolitics	strategic valuation discrepancy	requires geopolitical specification
Diplomacy	transformation cost versus gain	requires negotiated constraints
International relations	multiple state valuations	requires actor-specific utility maps
Warfare economics	cost-adjusted advantage Π	requires operational models and ethics

10 Proofs in Compressed Form

Proposition 1. If $fg = c^4$ and $f, g > 0$, then there exists a real h such that $f = c^2 e^h$ and $g = c^2 e^{-h}$.

Proof. Define $h = \log(f/c^2)$. Then $f = c^2 e^h$, while $g = c^4/f = c^2 e^{-h}$. □

Proposition 2. If $fg = c^4$ and $m \le f + g \le M$, then $|f - g| \le \sqrt{M^2 - 4c^4}$ on the real branch for which the square root is defined.

Proof. From $(f - g)^2 = (f + g)^2 - 4fg = (f + g)^2 - 4c^4$ and $f + g \le M$. □

Proposition 3. If valuation switching is free and both directions are admissible, global no-arbitrage implies $f = g = c^2$.

Proof. Free identity switching gives the two gains $g - f$ and $f - g$. Both must be nonpositive, so $f = g$. The product constraint then gives $f^2 = c^4$, and the positive branch gives $f = g = c^2$. □

11 Limits and Research Program

The combined framework is mathematically stronger as a representation than as an empirical theory. Several open problems remain: multidimensional fold calculus; q-adic or non-dyadic generalizations; full function-space characterizations of fold regularity; minimax theory for constrained tent estimators; conditions for existence of fold derivatives; and rigorous tests of whether fold-based features add predictive power beyond established bases and kernels [3]. On the arbitrage side, a full homology theory, continuous-state cycle detection, and empirical identification of transformation costs remain open [5].

For empirical work, the natural program is to specify observables for f, g, h, transformations T, and costs $C(T)$; estimate h under exact product constraints; test the sum bound; estimate fold-complexity and fold-regularity; and finally test whether measured transformation gains exceed costs out of sample. This separates mathematical identity, statistical estimation, and economic falsification.

12 Conclusion

The combined construction has a simple core. The multiplicative restriction reduces the two functions to one latent field,

$$f = c^2 e^h, \qquad g = c^2 e^{-h}. \tag{49}$$

The additive restriction bounds that field and therefore bounds the valuation discrepancy. Stronger Blancmange Analysis supplies the appropriate dyadic geometry, localized Faber–Schauder approximation, fold derivatives, and fold-regular learning. Arbitrage Geometry then interprets the same discrepancy as a potentially exploitable valuation gap and supplies transformation costs, cycle curvature, and global no-arbitrage structure. The result is a compact synthesis in which one latent field links reciprocal analysis, multiscale roughness, constrained estimation, and structured misalignment.

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Glossary

Arbitrage functional
Cost-adjusted valuation gain under an admissible transformation.

Arbitrage curvature
Net arbitrage accumulated around a closed transformation path.

Blancmange function
Takagi function generated by recursive dyadic tents.

Fold address
Binary sequence identifying nested dyadic intervals.

Fold derivative
A derivative-like limit comparing increments of a function with increments of a fold observable.

Fold metric
A metric combining Euclidean distance with distance along a Takagi–Landsberg graph.

Fold complexity
Weighted magnitude of localized dyadic coefficients across scales.

Faber–Schauder atom
A localized dyadic hat function used for continuous piecewise-linear approximation.

Latent field
The scalar field h that generates the reciprocal pair $f = c^2 e^h$ and $g = c^2 e^{-h}$.

No-arbitrage equilibrium
A state from which no admissible positive arbitrage transformation exists.

Valuation discrepancy
The difference $f - g$ between two valuation systems.

The End